

11

Acquisition of the syntax–discourse and semantics–pragmatics interfaces

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11.1 Where is the syntax–discourse interface?

In our journey through SLA, we have discussed the acquisition of morphology, syntax, lexicon, and the syntax–semantics interface. A common aspect of these properties is that they are considered to be language-internal, that is, regulated by rules intrinsic to the language faculty. As the diagram (Figure 11.1) based on White (2009) visualizes, the central box containing the core grammar modules: lexicon, phonology, morphology, syntax, and semantics, interfaces with meanings and sounds through dedicated interfaces (Logical Form and Phonetic Form). In this chapter, we will be thinking about properties of the grammar that touch and are contingent on the extralinguistic context, the life situation surrounding the language expressions. The

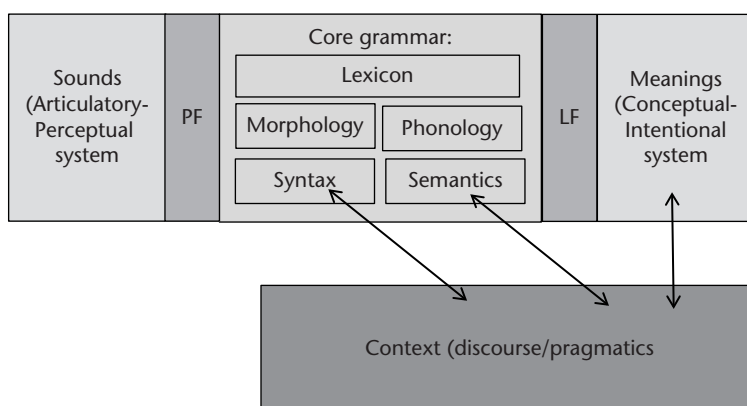


Figure 11.1 A working language faculty representation, based on White (2009)

meanings that we will encounter cannot be successfully computed, or appreciated, if context is not taken into account. Consequently, they are considered harder to figure out than other meanings and forms. As Figure 11.1 shows, context can be thought of as being outside of the box of the linguistic modules proper but certainly interacting with meanings; context has an effect on syntactic expressions as well.

To elucidate the terminology and combat confusion, we should start from the basics. Interpretation in language arises from the interaction of semantics and pragmatics. Sentences are built up in the morphosyntax, or the core syntax, module. Next, their propositional content is calculated in the semantics module; at the end of that process they are assigned truth conditions. The usage of the propositional content (i.e., truth conditions) in concrete communication is governed by pragmatic norms. In uttering a sentence, not only do we care about its truth conditional content, but we also aim at being appropriate, relevant, and cooperative, as the philosopher of language Paul Grice proposed in his famous lectures (Grice 1989). Therefore, meaning scholars often repeat the adage: Semantics proposes, pragmatics disposes. To understand exactly how semantics and pragmatics interact in the computation of the final meaning (what is said but also what is meant), semantics and pragmatics scholars battle it out with concrete proposals. The division of labor seems to be different for different constructions.

There has been a general confusion as to what linguistic properties belong to the semantics–pragmatics interface and which belong to the syntax–discourse interface. Furthermore, many linguists use these terms interchangeably, which can be confusing. I shall try to straighten out the terms

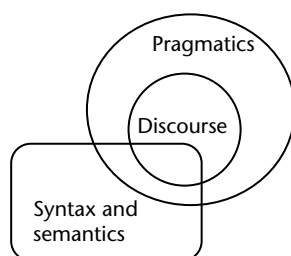


Figure 11.2 Relationship between syntax, discourse, and pragmatics

as much as I can. Although both can be considered external interfaces, the syntax–discourse interface properties represent the set while the semantics–pragmatics properties represent the superset (see Figure 11.2). The former include all constructions whose meaning computation and acceptability depend on information coming from the previous discourse: for example, what entity was mentioned in the previous sentence as given information (Topic) and information about what entity was required (Focus).¹ As we shall see below, properties that are calculated at this interface include preverbal and postverbal subjects in languages like Spanish and Italian, the use of overt subjects in Topic shift contexts, binding of pronouns, doubling of topicalized objects with clitics in, for example, Spanish and Bulgarian, contrastive focus, discourse-related word order optionality (scrambling) in German, etc. When definiteness is based on previous mention of a referent, its calculation is also discourse-based.

The level of linguistic pragmatics is much larger in scope than discourse–pragmatics. The *Handbook of Pragmatics* edited by Horn and Ward (2004) lists conversational implicature, presupposition, reference, deixis, definiteness, and speech acts as the principal domains of pragmatics. Of these six large areas, only reference and partially definiteness are contingent on the discourse context, while properties in the other areas are interpreted depending on knowledge of the world, Gricean maxims of co-operation, and other universal pragmatic principles. Speech acts are both culture- and education-dependent. When talking about the semantics–pragmatics interface (not including discourse-based constructions), linguists usually have in mind

¹ Semantics is not bypassed by this interface, as its name might suggest. A more appropriate name would be syntax–semantics–discourse interface, as the linguistic structures produced in the core syntax are checked for their truth conditions as well as their appropriateness to the discourse context.

scalar implicatures, presuppositions, deictic expressions, aspectual and other types of coercion, transfer of reference, and so on. Roughly, this interface regulates the difference between what is said (literally) and what is actually meant and understood. Universal pragmatic principles as well as knowledge of the world are crucial for comprehending language and can be transferred from the native language. Thus the semantics–pragmatics interface is not very likely to present much difficulty to nonnative speakers, unless some computations with added cognitive complexity are involved. Figure 11.2 represents this view of the relationship between syntax, semantics, pragmatics, and discourse.

It would be more *precise* if we used “syntax–discourse” for the interface where we calculate meaning based on our knowledge of the discourse context while reserving “semantics–pragmatics” for the interface where we recalculate meaning coming from the semantics (propositional content of sentences) based on knowledge of the world and universal pragmatic principles. Furthermore, since the two terms are in a set–superset relation, it is not a contradiction to use the wider term standing for the narrow term. When we use the narrow term (syntax–discourse), we imply that the wider term (semantics–pragmatics) is not applicable. This usage itself is a scalar implicature, as we shall see later in the chapter. Of course, we should always keep in mind that for calculation of the complete meaning, all modules and interfaces have to work in consort.

11.2 Marking of Topic and Focus across second languages

The most prominent and frequent use of discourse information in language is the marking of Topic and Focus, known as Information Structure. This structure regulates the information available to the speaker and hearer when they encode and decode a sentence in context. If a constituent has already been mentioned (activated) in the discourse or is well known to speaker and hearer, it acts as Topic. If a constituent encodes new information not previously known or mentioned, it is Focus. Questions open the possibility to answer with new information (e.g., *When did she arrive? Last night.*) Languages employ very different means of signaling this all-important discourse information, that is, marking Topic and Focus. The main means are permutations on neutral word order (left and right dislocations), doubling of dislocated elements, other word order changes, and specific

intonation. The L2A challenge is when learners have to acquire a different way of marking Topic and Focus from their native system marking.

In this section, I will start by presenting studies investigating clitic-doubling as a means of signaling topicalization. Recall that a clitic is a special type of pronoun which is phonologically dependent on a host and may double (refer to) an argument already present in the sentence. There has been quite a lot of work recently on this particular property of clitic doubling. For starters, let me note that topicalization, that is, moving an object to sentence-initial position to mark it as old information, is common across languages. However, languages differ in whether the topicalized constituent is doubled with a clitic referring to it. Note the two examples below from English (1) and Bulgarian (2).

(1) A: Did Jane like the wine?

B: Oh, the wine she didn't drink (*it). She stuck to lemon ices.

(2) A: Maria haresa li vinoto?

Maria like-PAST Q wine-DET

'Did Maria like the wine?'

B: Vinoto tja ne *(go) pi. Boleshe ja glavata.

wine-DET she not it drink-PAST hurt her head

'The wine she didn't drink. She had a headache.'

Note that in the same context, both English and Bulgarian can (optionally) dislocate the object *the wine* because it is mentioned in the question and thus old information. However, in the English sentence in (1B), the pronoun *it* referring to the wine is ungrammatical, which is marked by the star inside the parentheses (*it* cannot be in that position). On the other hand, the Bulgarian sentence in (2B) sounds strange without the clitic *go* referring to and agreeing in gender with *vinoto* 'the wine', which is marked by the star outside the parentheses (meaning that *go* is not optional). Note that SVO is the neutral word order in both English and Bulgarian and the object dislocation cannot be uttered out of the blue, without appropriate context. In acquiring the construction, English-native learners of Bulgarian have to add the clitic to their mental representation; in the opposite direction, learners have to acquire the ungrammaticality of the pronoun. We shall see examples of both types of studies below.

One of the first studies investigating the acceptance of topicalization, known as clitic left dislocation in Spanish, was Valenzuela (2005, 2006). Valenzuela capitalized on the observation that in Spanish, only specific objects as in (3ab) are successfully doubled with a clitic, while generic objects as in (4ab) are not. The examples below are from her sentence selection task. In this task, near-native learners of Spanish heard a short context story in Spanish where the discourse antecedents were introduced. They then heard two possible concluding sentences for the story in Spanish and were asked to select the most appropriate answer. They could also choose ‘neither a nor b’ as well as ‘both a and b’, as shown:

- (3) Ayer por la mañana, Eva se fue a la universidad y vio a su amigo Pedro y a su amiga Inés, pero como tenía muchas cosas que hacer . . .
‘Yesterday morning, Eva went to the university and saw her friend Pedro and her friend Inés, but since she had many things to do . . .’
- a. A Pedro, no lo saludó. ← expected response
Acc. Pedro, not CL greeted
‘Pedro, I didn’t greet.’
- b. A Pedro, no saludó.
Acc. Pedro, not greeted
‘Pedro, I didn’t greet.’
- c. Ni a ni b
‘Neither a nor b’
- d. Ambas a y b
‘Both a and b’
- (4) El médico de Paco le dijo que tenía que tomar mucha agua. También le dijo que tenía que comer más fruta y verdura. Pero Paco es muy cabezón y . . .
‘Paco’s doctor told him that he has to drink a lot of water. He also told him that he needs to eat fruit and vegetables. But Paco is very stubborn and . . .’
- a. Agua, no toma nunca. ← expected response
water, not drinks never
‘Water, he never drinks.’

- b. Agua, no la toma nunca.
 water, not CL drinks never
 'Water, he never drinks.'
- c. Ni a ni b
 'Neither a nor b'
- d. Ambas a y b
 'Both a and b'

Valenzuela's near-native participants were 100% accurate on the sentences as in (3), even more accurate than the native speakers. On the other hand, with generic dislocated objects, the Spanish native speakers gave the expected response (4a) only 53% of the time. They also chose the clitic-doubled option 14% of the time, and the "both constructions are possible" option 19% of the time. For the near-native speakers the respective answers were 40%, 37%, and 20%, which was significantly different from the natives. Valenzuela argued that the learners' knowledge of this construction is deficient.

Ivanov (2009, 2012) is another study that looked at the same property, but in the acquisition of Bulgarian. For the purposes of our comparison here, we have to know that Spanish and Bulgarian work similarly. The only difference is that Bulgarian topicalized objects have to be doubled with a clitic no matter whether they are dislocated or in their usual position after the verb. This possibility is observable in the examples from the test items. Ivanov tested native speakers of English who were advanced and intermediate learners of L2 Bulgarian. First of all, just as Valenzuela did, Ivanov ascertained that the research participants knew the syntactic properties of clitics. There is no need to search for knowledge of clitic doubling at the syntax–discourse interface if the learners have not acquired clitics in the first place. Secondly, he used an acceptability judgment task where the test items were presented aurally as well as in writing. Intonation is very important for the marking of Focus and Topic and test sentences should not be evaluated without the required intonation. A test sentence with the choices that had to be evaluated on a scale of 1 (unacceptable) to 5 (perfect) is given in (5). The number sign (#) means that the sentence is unacceptable in this context.

- (5) Q: Poluči? li koleta ot Peter
 receive-2.SG.PAST Q package from Peter
 'Did you receive the package from Peter?'

- A: a. Koleta **go** polučih minalata sedmica. [+Object fronting][+Clitic doubling]
 package CL.ACC receive-1.SG.PAST last week
 'I received that package last week'
- b. Minalata sedmica **go** polučih koleta. [-Object fronting][-Clitic doubling]
- c. #Koleta polučih minalata sedmica. [+Object fronting][-Clitic doubling]
- d. #Minalata sedmica polučih koleta. [-Object fronting][-Clitic doubling]

Ivanov tested both accusative and dative objects in Topic as well as Focus contexts. The exact same sentences are acceptable with clitic doubling in Topic contexts and without clitic doubling in Focus contexts as in (6). As I mentioned above, questions create a Focus context because the answer is new information.

- (6) Q: Kakvo zagub-i dokato bjaga-še тази сутрин?
 what lose-2SG.PAST while run-2SG.PAST this morning
 'What did you lose when you were running this morning?'
- A: a. #Ključovete za kolata gi zagubix. [+Object fronting][+clitic doubling]
 keys-DET for car-DET CL.ACC lost-1SG
 'I lost the car keys.'
- b. #Zagubix gi ključovete za kolata [-Object fronting][+clitic doubling]
- c. Zagubix ključovete za kolata. [-Object fronting][-clitic doubling]
- d. Ključovete za kolata zagubix. [+Object fronting][-clitic doubling]

While Ivanov's intermediate learners had not acquired the acceptability of these sentences in context, the advanced group ratings demonstrated a significant difference between the acceptable and unacceptable sentences. In addition, their ratings were not significantly different from the Bulgarian native speakers' ratings. Thus, it seems that the results of Valenzuela's and Ivanov's experiments point to different outcomes. In order to resolve this apparent contradiction, Slabakova and Ivanov (2011) reanalyzed Valenzuela's data on generic sentences as in (4). Conspicuously, the behavior of the native speakers on these sentences was far from categorical, with the expected answer being chosen only about half of the time. This suggests a variability in the input that the learners are exposed to, which may well have affected their own choices.

A second experimental study on Spanish clitic left dislocation, Slabakova, Kempchinsky, and Rothman (2012), attested successful acquisition of an

additional semantic constraint on the dislocation: the clitic-doubled and dislocated object (the chairs) was just a subset of the object mentioned in the context (furniture).

- (7) [context: What did the movers do with the furniture?]
 Las sillas las dejaron en el pasillo, pero no sé dónde están las mesas.
 the chairs CL left.3PL in the hallway, but NEG know.1SG where are
 the tables
 ‘The chairs, they left them in the hallway, but I don’t know what they
 did with the tables.’

Donaldson (2011a,b, 2012) examined clitic right dislocation and clitic left dislocation, among other properties, in near-native L2 French. French is a language that uses these constructions a lot. His study followed the natural conversations of ten native speaker–near-native speaker dyads. He demonstrated that the behavior of the near-native speakers was not qualitatively and quantitatively different from the native behavior.

Finally, Slabakova and García Mayo (2015) tested topicalizations as in (1), repeated here, contrastive fronted focus as in (8), and the so-called left dislocations as in (9), in English as a second and a third language.

- (1) A: Did Jane like the wine?
 B: Oh, the wine she didn’t drink (*it). She stuck to lemon ices.
- (8) Customer: Can I get something for breakfast? Maybe a bagel?
 Mark: No, sorry. We’re out of bagels. A bran muffin I can give you/*it
 to you.
- (9) My wonderful Felix, everyone likes *(him).

While topicalizations and fronted focus constructions depend on context and do not allow a pronoun that refers to the moved constituent (called a “resumptive” pronoun), the left dislocation can be uttered out of the blue, and the pronoun in object position refers to the sentence-initial phrase. That’s because that phrase did not move from object position but was adjoined to the sentence. The format of this experiment was very similar

to those of Ivanov and Slabakova et al's experiments, with a rating scale from 1 to 7. However, the findings were very surprising. Neither the second language learners nor the third language learners² showed any knowledge of topicalization, although they were better on focus and left dislocations. The measure of success in this study, as above, was whether the ratings of the unacceptable construction were significantly different from the ratings of the acceptable constructions. On topicalization, the ratings of all learner groups converged on essentially the same value, in the middle of the scale (3.5). That is, they were not sure about the acceptability of both the sentences with the pronouns and those without.

In summary, we see mixed results in the acquisition of dislocations, with some clear successes (Ivanov, Donaldson, Slabakova et al.) and some disasters (Slabakova and García Mayo). It is very tempting to use the Interface Hypothesis as an explanation of these findings. The Interface Hypothesis, proposed by Antonella Sorace and her associates, operationalized the idea that learners will show much more variability and optional behaviors at interfaces in general. This variable behavior can be due to limitations in working memory, processing capacity or processing efficiency that are experienced by bilinguals and multilinguals. In addition, inefficient resource allocation that obtains as a byproduct of bilingualism has been suggested as another possible reason for underperformance at interfaces (Sorace 2011). In essence, the idea is that when too much information has to be used in the calculation of some meaning or some word order, the bilingual or multilingual mind experiences strain of capacity. This strain is manifested in acceptance (or indiscriminate ratings) of two opposing constructions, for example, left dislocations with and without a clitic/pronoun doubling, as described in the Slabakova and García Mayo experiment above. This optionality is often reflected in significantly different ratings of native and nonnative experimental participants. However, see Exercise 11.1 on the significance of those differences.

Over time, the Interface hypothesis was extended to virtually all contexts and consequences of bilingualism (e.g., adult L2 acquisition, child 2L1, child L2, and L1 attrition, see Sorace, 2011, Tsimpli et al. 2004). In part based on the recent emergence of quality data sets, examining multiple languages and multiple interface-conditioned domains of grammar, the

² Who were Basque–Spanish and Spanish–Basque bilinguals learning English as a third language.

Interface Hypothesis has been revised (or clarified, see Sorace, 2011) to no longer make claims about vulnerability for all interfaces. At present, the IH targets (emerging) optionality at the external interfaces only and at the level of near-native ultimate attainment or first language attrition (i.e., it is not part of development). However, other researchers (e.g., Lardiere 2011 and White 2011) argue that a linguistic issue manifested at near-native levels of proficiency should also be noticeable when learners are less proficient. After all, progression in L2 proficiency entails better grammar knowledge and better computational efficiency at the later stages than at the earlier stages.

But why would syntax–discourse interface properties be more vulnerable than other interface properties? The problem is hypothesized to stem from the very limitations in cognitive resources in bilinguals, but it is noticeable in child language comprehension as well. When faced with a decision about meaning that depends on juggling several sources of information (the syntax, the extralinguistic information, and frequently intonation as well), children and inexperienced L2 learners might suffer a breakdown in meaning calculation, at which point they resort to guessing. Reinhart (2006) specifically argues for such a scenario in scalar implicature and pronoun comprehension by children, but it is easy to see how this explanation can be extended to second language learners. In such learning situations, experimental results frequently hover around 50% accuracy. Children of ages 4 to 6 may still be developing the brain capacity to manage several sources of information. That same capacity may be taxed by the costs of inhibition needed to deactivate the multiple grammars present in a multilingual’s mind.

While the Interface Hypothesis is rigorously argued, based on a principled distinction within the language faculty and supported by a number of studies, there are also other studies whose results point against it. In this section, we only looked at object fronting to mark Topic, and we reviewed more studies that documented success than studies that attested failure of learning. For example, in acquiring clitic left dislocations in Spanish, advanced learners were indistinguishable from native speakers (Slabakova et al. 2012). In the opposite learning direction, from Spanish to English, even very advanced learners were at a complete loss with the same construction (Slabakova and García Mayo 2015). Obviously, this discrepancy cannot be due to the uniform difficulty of the syntax–discourse interface. There must be some other factor, or factors, involved. Slabakova (2015c) attributes the difference in success rates in the two learning directions to the

frequency of the two constructions in the target languages. In English, topicalization is very rare while clitic left dislocation in Spanish is about 1000 times more frequent, according to corpus data (Slabakova 2015c).³ In addition, English does have a construction such as left dislocation (see example (9) above), which not only allows but requires a resumptive pronoun. As we have seen previously in this textbook, quantity and quality of input may provide the best explanation again: it is only rarely that English learners experience topicalization, and the competition from left dislocation may make the learning task even more confusing. After all, the meanings of the two constructions overlap even if their distribution does not.

While the Interface Hypothesis is falsified by such results, it still points to a valid source of difficulty, as we will see in the next sections. It is still the case that constructions at the syntax–discourse interface are difficult, it is just not the case that they are uniformly difficult, and also not the case that such difficulty is insurmountable.

11.3 Word order: constraints and strategies

In this section, we will look at some studies that investigate the order of subject and verb in discourse-dependent contexts. As you will see in the examples below, in (neutral SVO) languages that allow this, the subject can follow the verb if it is new information, or focus. In fact, this construction is part of the cluster of constructions that rely on the Null Subject Parameter. In other words, the possibility of null subjects and Verb–Subject order are related (Rizzi 1982, Belletti 2000, 2003). But first, let us distinguish between contrastive and presentational focus. Although both present new information, contrastive focus necessarily compares two propositions as in (10), where caps stand for emphatic intonation.

(10) Q: Do you like my new jacket?

A: I like your new SHIRT; the jacket not so much.

³ The data for this comparison come from the Switchboard Telephone Speech Corpus in English, described by Gregory and Michaelis (2001), and from the NOCANDO corpus in Spanish, Brunetti, Bott, Costa, and Vallduví (2011).

Presentational focus, on the other hand, does not evaluate options but simply introduces new information (11). It is marked *in situ*, which means that nothing has moved about in the sentence (É. Kiss 1998, Rochemont 1998, Domínguez 2004).

(11) Q: Who came late last night?

A: Linda came late.

A third term we shall make use of here is “thetic sentences”: these are produced out of the blue, with no previous mention, or they are answers to the question “What happened?” as in (12).

(12) Q: What happened at the party last night?

A: The police came. We were making too much noise.

Observe that the word order in English is Subject–Verb in all of the answers above. It is possible for English to invert the subject and the verb, but this happens rarely and only under very special conditions; it is the (first) auxiliary that moves over the subject:

(13) Never have I heard such an amazing proposal.

But of course, English is not a null-subject language. Spanish, a null-subject language, allows the subject to follow the verb. Inversion is possible even in unfocused (out of the blue) contexts. However, native speakers invert the subject much more with unaccusative than with unergative verbs.

Notice: this is a way in which the discourse, the lexicon, and the syntax interact in complex ways! Can you think of other cases of such interactions?

Recall from Chapter 9 that in unaccusative verbs such as *arrive*, the subject has the semantic role of Patient, or Theme, while in unergative verbs such as *shout*, the subject is an Agent. We reviewed proposals to the effect that the two types of subjects start from different underlying positions in the verb phrase, which is reflected in the inversion preferences. Examples are from Lozano (2006). In (14) and (15), the lead question, which asks about the whole event, makes the answer all new, and therefore thetic.

- (14) Q: ¿Qué pasó anoche en la fiesta?
‘What happened last night at the party?’
A: Vino la policía. unaccusative: preference for VS
‘The police arrived.’
- (15) Q: ¿Qué pasó anoche en la calle?
‘What happened last night in the street?’
A: Una mujer gritó. unergative: preference for SV
‘A woman shouted.’

However, when the lead question asks about a specific person and the answer involves presentational focus, the judgments are different: both unergative and unaccusative verbs sound better with the inverted VS order. In other words, presentational focus neutralizes the unaccusative–unergative distinction.

- (16) Q: ¿Quién gritó anoche en la calle?
‘Who shouted last night in the street?’
A: Gritó una mujer. unergative verb: VS
‘A woman shouted.’
- (17) Q: ¿Quién vino anoche a la fiesta?
‘Who arrived last night at the party?’
A: Vino la policía. unaccusative: VS
‘The police arrived.’

In order to master the complete data set in (14)–(17) above, a learner of Spanish has to be aware of the possibility and preference for VS structures with unaccusative verbs but not with unergative verbs. This is lexical knowledge reflected at the lexicon–syntax interface (we saw some studies on this issue in Chapter 9). In addition, sensitivity to the discourse status of the answer sentence (out of the blue/thetic or presentational focus) should be acquired. That sensitivity is arguably reflected at the syntax–discourse interface. Since this is an attractive paradigm to investigate, it has provoked considerable research interest. I shall mention a few studies here. Hertel’s (2003) study investigated the production of presentational focus VS order in L2 Spanish by American English natives at four proficiency levels (beginner, low intermediate, high intermediate, advanced). She found that her

learners did favor SV with unergatives but VS with unaccusatives and, therefore, were sensitive to the syntactic effects of the lexicon even at advanced stages of acquisition. Unexpectedly, she also found that her participants were producing presentational focus at even lower proficiency levels than advanced.

Lozano (2006) created an experimental study where learners of Spanish with English and Greek as native languages evaluated the acceptability of answers such as those in (14) to (17) in unfocused and presentational focus contexts. Greek as a native language is investigated in this study because Greek presentational focus only allows SV sentences (just as English) while similar to Spanish, it inverts unaccusative VS orders. Thus, Greek learners have a possible transfer advantage, but only for the word orders in unfocused contexts. In the experiment, even the English learners of Spanish were accurate with the lexical property. Greek and English learners demonstrated significant preference for SV with unergative verbs and VS with unaccusative verbs. However, the two learner groups, which were at advanced proficiency, allowed both word orders (SV/VS) in presentational focus contexts equally. In other words, they did not feel that VS was more appropriate in such contexts. Learners were not sensitive to the discursive constraint although they were able to observe the lexical distinction. Lozano argued that his findings support the Interface Hypothesis, as they clearly do.

Later work by Lozano and Mendikoetxea (2010) looked at the production of VS structures in a L1 Spanish–L2 English learner corpus and a comparable English native corpus. They classified the productions as obeying the interface principles: inversions are allowed when the verb is unaccusative, when the subject is heavy (contains many words) and when it is focused. Learners and native speakers obeyed all these principles equally well. The errors that learners made were in the syntactic form of the sentences with inversion. They used the expletive *it* or nothing instead of *there* in the preverbal position:

- (18) a. *In the name of religion **it** had occurred many important events . . .
 b. *It is difficult that Ø exist volunteers with such a feeling against it.

Focus can be marked by subject–verb inversion, but it can also be marked by the appropriate emphatic intonation (*I want to see YOU*) or by a cleft construction (*It is you that I want to see*). Belletti (2013) calls these various

ways of encoding Focus “different answering strategies” because they are all legitimate in Italian and in many other languages. However, native speakers and L2 learners exhibit striking differences in the preference for these answering strategies, with most learners maintaining their native answering strategies in the L2 (Belletti and Leonini 2004, Belletti, Bennati, and Sorace 2007, Lozano 2006). For example, French learners of Italian employ mostly clefts, while English learners of Italian employ intonation marking by stressing the subject. Studies in the other learning direction, Italian to English (Belletti 2013), suggest that learners respect grammatical necessities and only employ native answering preferences when the grammar allows it. Even when demonstrating that the null subject property of the target L2 language has been properly (re)set, the free VS inversion option is accessed only to a limited extent. Belletti argues that the choice of a particular strategy is only partly conditioned by grammatical factors.

Arguments and findings such as Belletti’s bring up the question of the representational versus processing source of difficulties and optionality at the syntax–discourse interface. They suggest that even if nativelylike grammatical representations are in place, and especially when the grammar allows choices, even near-native learners will reach for the native answering strategies under the burden of communication and bilingualism pressures. Of course, I am talking of preferences for options that are allowed by the target grammar, and the near-native learners discussed by Belletti are not choosing anything unacceptable. On the other hand, utilizing nonnative strategies does not exclude having grammatical problems, too, for the less experienced learners.

11.4 Pronoun reference

In the previous section, we discussed how the inversion of the subject and the verb, VS word order, can signal the presentation of new information in some languages. A similar interaction of grammar and discourse–pragmatics is found in pronoun reference calculation. Postverbal subjects are part of the Null Subject Parameter cluster, and thus are related to another interesting manifestation of discourse sensitivity: a division of labor between null and overt pronominal subjects for different communication situations. In a nutshell, native speakers of Spanish and Italian prefer to use null subjects when the subject of the main clause is maintained as the subject

of the embedded clause. This is known as Topic maintenance. On the other hand, they prefer to use overt pronominal subjects when the embedded subject refers to another argument in the main clause, but not the subject. This is known as Topic shift.⁴ Let me illustrate with a couple of sentences from Italian (from Belletti, Bennati, and Sorace 2007).

- (19) La vecchietta saluta la ragazza quando Ø attraversa la strada.
 the old.lady greets the girl when crosses the street
 ‘The old lady greets the girl when she crosses the street.’
 she=the old lady
- (20) La vecchietta saluta la ragazza quando lei attraversa la strada.
 the old lady greets the girl when she crosses the street
 ‘The old lady greets the girl when she crosses the street.’
 she=the girl

In processing sentences such as (19) and (20), (Belletti et al. 2007) found that Italian near-native speakers interpreted the overt pronominal subject of the embedded clause as referring to the lexical subject of the main clause 30% of the time, while the natives only interpreted it in this way 5% of the time; a significant difference. At the same time, 65% of near-native answers and 85% of native answers converged on the correct interpretation (embedded subject refers to matrix complement). Thus, the Italian near-natives in this study were less sensitive than the native speakers to Topic shift discourse situations. In addition to being statistically significant, this lower sensitivity is considered an important competence distinction, because the near-native speakers of Italian were very proficient in all other respects. See Exercise 11.1 for another interpretation of these findings.

Since this division of labor of null and overt pronouns may be a processing strategy, Keating, VanPatten, and Jegerski (2011) tested three groups of speakers in a processing experiment. One group contained monolingually raised native speakers of Spanish, the second group contained heritage speakers of Spanish, and the third group, adult L2 learners. They saw sentences like (21) on a computer screen and had to choose one of the two answers (a forced choice task).

⁴ Carminatti (2002) postulated a division of labor between overt and null pronouns in the embedded clause. She called it the Position of Antecedent strategy and argued that it is a processing constraint.

- (21) Daniel ya no ve a Miguel desde que Ø se casó.
'Daniel no longer sees Miguel ever since he got married.'
¿Quién se casó ?
'Who got married?'
A. Daniel B. Miguel

Although the results of the Keating et al. (2011) study demonstrate a statistically significant bias in antecedent assignment, they did not reveal a neat division of labor between null and overt pronouns overall. The monolingually raised Spanish speakers did prefer the main subject as the antecedent of the null pronoun in the embedded subject position (so they answered *Daniel* in (21)), but their preferences for the overt pronoun were rather more individually varied and noncategorical as a group. The heritage and the L2 learner groups did not demonstrate significant preferences for one or the other pronoun reference. The authors argued that there are significant distinctions between Spanish and Italian with respect to the antecedent strategy (i.e., the null pronoun refers to main clause subject, *Daniel* in this example; overt pronoun refers to main clause object, *Miguel* in this example). While in Italian this strategy is obeyed much more categorically, in Spanish it is not. As a result, the learners may not have had sufficient input to acquire the preference, while the heritage speakers may have lost it under the influence of their English. The lack of clear antecedent biases for either overt or null pronoun suggests that the L2 learners are still sorting out the pragmatic constraints on pronominal subject distribution, and for them the null and overt pronouns are in completely free variation. However, another possibility arises with respect to the heritage speakers. Shin and Cairns (2009) found that sensitivity to Topic shift contexts emerges in Mexican children between the ages of 7 and 9 years. Since the heritage speakers in this study (and generally in the US) are exposed to English and become bilingual before the age of 7, it could be that they never acquired the strategy.⁵ In any event, the results of Keating et al., both for heritage speakers and for L2 learners, are in line with explanations of problems at the syntax–discourse interface.

Finally, Zhao's (2012) study provides a very different perspective on the same issue at this interface. Chinese is a null-subject language, but it also

⁵ This would be a case of incomplete native language acquisition, as discussed in Chapter 5, see Montrul (2008).

allows null objects. It is interesting that, in contrast to Romance languages, both the embedded null subject and the embedded overt subject can refer to either a local antecedent (such as the main clause subject *John* in (22)) or an external antecedent (someone mentioned in the discourse but not in the same sentence). This is what the subscript “i/j” stands for: “i” is the subscript of *John* and *he*, while the referent of “j” is not explicitly mentioned but known from the discourse. The following examples will help you understand the Chinese facts:

(22) *John_i says that he_{i/j} likes Tom.*

(23) *Xiao Zhang_i shuo Ø_{ij}/ta_{ij} xihuan Lao Wang.*
Xiao Zhang say he like Lao Wang
 ‘Xiao Zhang says that he likes Lao Wang.’

Now compare the two sentences in (24) and (25). In the English sentence, the object pronoun *him* can refer to John and to a third person unmentioned in this particular sentence, but known to the interlocutors. In the Chinese sentence, the overt pronoun *ta* has the same interpretation as the English *him*. However, the null object pronoun Ø_{*ij} can only refer to the third person familiar from the discourse, not to Xiao Zhang.

(24) *John says that Tom likes him_{i/j}.*

(25) *Xiao Zhang_i shuo Lao Wang xihuan Ø_{*ij}/ta_{ij}.*
Xiao Zhang say Lao Wang like him
 ‘Xiao Zhang says that Lao Wang likes him.’

As you can appreciate, this is quite a complex set of data for the learners of Chinese to acquire, given that their native English allows no null pronouns, either as subject or as objects. Following up on Yuan (1993), which had established that learners of Chinese accept null subjects and null objects from very early stages of acquisition, Zhao’s study asked if learners actually had the correct interpretation for the null pronouns they accepted. She used a picture selection task with two types of pictures. In one type of picture, a speaker was talking about a third person; in the second type of picture, the speaker was referring to himself. These two types of pictures try to capture the two possible interpretations of *John_i says that he_{i/j} likes watching*

football. Learners had to rate four sentences with respect to each picture. The prediction was that, even if they had null elements in their L2 grammar, learners of Chinese might not be able to make the interpretive distinction between the null embedded subject and the null embedded object. To be fully competent, they should rate the sentences low where the embedded object is null but the picture shows it refers to the matrix subject (see example (25)). This is exactly the state of knowledge that Zhao uncovered, at least with the advanced learners. Their ratings patterned with the native speakers'. The intermediate learners' ratings were somewhat shakier, but still largely in line with the natives. All in all, these results are a success story.

Why should the Chinese syntax–discourse interface be different from the Romance syntax–discourse interface? While we need more studies with various language pairs, it is already clear from the research summarized in this section that the syntax–discourse interface is not monolithic, and that even very similar properties across various languages may create great difficulties for learners, or not so much difficulty. This situation calls for the scrutinizing of other factors that can impact behavior at this interface, such as frequency of the constructions in the input, processing strategies, and the relationship of discourse properties with other parts of the grammar.

11.5 Intonation at the syntax–discourse interface

Since intonation is one of the basic means of signaling information structure, this chapter would be remiss if it did not touch on this issue, although not much work has been done to date, at least in L2 acquisition. Please look at Figure 11.1 again. This research would be investigating the missing arrow between the discourse–pragmatics box and phonology, within the core grammar. But first of all let us recall the definition of prosody. “Prosody refers to the way in which words are grouped in speech, the relative acoustic prominence of words, and the overall tune of an utterance” (Breen et al. 2010: 1047). Prosody involves acoustic features such as fundamental frequency (F0), duration, and loudness of the speech unit. The combinations of these acoustic features give rise to psychological perceptions such as phrasing (grouping of words), stress (prominence), and tonal movement (intonation). In every sentence, there is one word in particular that receives main prominence relative to its neighbors. Such rhythmic prominence is

known as Nuclear Stress (NS). In many languages, NS serves the important function of identifying the focus domain of a sentence.⁶

In a series of experiments, Breen et al. (2010) asked their English-speaking research participants to produce sentences with focus, having manipulated the Information Structure by providing preceding context. They analyzed the recordings and found that speakers systematically provided prosodic cues as to which part of the sentence is focused material. They also provided clues as to whether the whole sentence (wide focus in *thetic* sentences) or only parts of the sentence was new information (narrow or presentational focus). This would be the difference between *What happened this morning?* versus *What did Joe fry this morning?* Finally, they established that speakers *can* distinguish between contrastive and presentational focus, but don't do it all the time. English native speakers signal contrastive focus only when they perceive syntactic ambiguity and want to disambiguate the message (*JOE fried an omelet, not Stan.*)

The acquisition study we examine here is by Nava and Zubizarreta (2009, 2010). It is based on the important theoretical treatment for Spanish by Zubizarreta (1998). The authors start by noting that while in English the primary mode of signaling information structure (Topic and Focus) is prosody, Spanish uses word order flexibility for the same purpose.⁷ More specifically, the distinction between English and Spanish is the variable placement of the nuclear stress (NS). In English each constituent can be emphatically stressed, and hence the NS can be on any constituent, while in Spanish only the last constituent can be emphatically stressed, that is, the NS is always last in the sentence. Underlining in the following examples marks NS.

When a sentence is an answer to a *What happened?* question, or it is uttered out of the blue, the NS rule is the unmarked stress pattern. The last word of the sentence gets the accent.

- (26) a. What happened? b. A boy broke his leg. (wide focus)
 c. Barbara drew pictures on the covers.

⁶ Chomsky and Halle (1968), Chomsky (1971), Jackendoff (1972), Ladd (1996), Zubizarreta (1998).

⁷ See also Ladd (1996).

English may also generate NS on the subject or NS on the verb. This variable NS is exploited to distinguish the wide focus statements as in (26) from narrow focus statements as in (27).

- (27) a. Who broke his leg? b. A boy broke his leg. (narrow focus)

Another important prosodic rule of English is Anaphoric Deaccenting. This is the rule postulating that already known material loses prominence and the NS shifts to its sister node. In (28), since stamps have already been mentioned, the stress shifts to the verb *collect*. Pronounce the sentence with the emphasis on *stamps*, and appreciate that it sounds subtly wrong. Compare that with the Spanish equivalent in (29). If you speak Spanish, try to appreciate the prosodic distinction between (28) and (29).

- (28) Q: Why are you buying that old stamp? A: Because I collect stamps.

- (29) Q: ¿Por qué compras ese sello tan viejo? A: Porque colecciono sellos.
why are-you-buying this stamp so old because I-collect stamps
'Why are you buying this old stamp?' 'Because I collect stamps.'

Deaccenting is a rule that applies after the NS rule: they have to come in a sequence. Nava and Zubizarreta predicted that learners of English would acquire the deaccenting rule before the NS rule, because Spanish does not have deaccenting. Thus, this rule is completely new to the learners and without an equivalent in their native grammar. The native NS rule, on the other hand, has to be restructured in the L2. In a sense, they predicted it would be easier to learn a new rule than to change a previously existing native rule.

In their experimental study, Nava and Zubizarreta tested the NS assignment rule and the deaccenting rule. They found that quite a big proportion of their intermediate and even advanced learners did not obey the English prosodic rules. However, they also found learners who had acquired both rules. Looking at individual performances, they established that the deaccenting rule came in the grammar more easily than restructuring the NS rule. Ten learners had the former rule only, and none had the latter rule only. Thus their prediction was supported on the individual level.

Zubizarreta and Nava (2011) compared their results above with the results of Lozano (2006), which we already discussed. The comparison of

the two learning directions (Spanish-to-English and English-to-Spanish) shows that the difficulty lies not only in learning the way of marking focus in the respective target language, but also generating the grammatical structures that are necessary to accomplish this marking. In other words, prosody always works in consort with grammar, and clearly complicates the acquisition picture.

11.6 Scalar implicatures

In this section, we tackle an issue that is arguably at the interface of semantics and pragmatics: scalar implicatures. Recall that very few properties at the semantics–pragmatics interface have been investigated in L2 acquisition. There is a reason for this: pragmatics is arguably largely universal. While some differences do exist, they are notably in the speech act area of pragmatics: the way we make a compliment or ask for a favor is not the same across cultures. Since this work is not done within the generative framework, we will not review it here, but see other recent surveys.⁸ To exemplify the universality of pragmatics, let's consider space and temporal deixis. Deixis refers to the phenomenon wherein understanding the meaning of certain words and phrases in an utterance requires contextual information. Words that have a fixed semantic meaning but have a denotational meaning that constantly changes depending on time and/or place, are deictic. Classical examples involve the meaning of personal pronouns and adverbs such as *tomorrow* and *here*.

- (30) a. We went to the cinema last night.
b. We live longer than men.

If she knows the meaning of the pronoun *we*, the L2 speaker of English certainly understands that the group of people referred to in (30b) is “all women,” while the referents of *we* in (30a) are contextually determined (family, friends, partners, etc.).⁹

⁸ See Bardovi-Harlig (2011, 2013) for two evaluative surveys of interlanguage pragmatics.

⁹ For a review of work on all areas of linguistic pragmatics, see Slabakova (2013).

This type of implied meaning, in this case reference, is part of almost any communication. Remember, we are looking at the difference between what is said and what is meant. Consider the following example of a well-known pragmatic inference:

(31) Some professors are smart.

Most people would agree that, in hearing the utterance in (31), they understand that the speaker has conveyed the assumption in (32).

(32) Not all professors are smart.

Notice that (32) is not encoded by the speaker's utterance, nor is it part of what the speaker has *said*. Rather, (32) is an assumption inferentially derived by the hearer on the basis of what the speaker has said. Logically speaking, *some* means *some and possibly all*. But if the speaker of (32) had meant *all professors are smart*, she would have uttered (33) or (34), being maximally informative, and not (31). Since she didn't, then we can safely assume she means (32).

(33) Professors are smart.

(34) All professors are smart.

The first systematic attempt to explain how the inference in (31) is derived is due to the philosopher of language Paul Grice. In a series of lectures presented at Harvard in 1967, published later as Grice (1989), he offered a comprehensive framework of the mechanisms of inferential communication. More specifically, he suggested that communication is essentially governed by certain rational expectations about how a conversational exchange should be conducted, which he called "maxims." According to Grice's five maxims, interlocutors are normally expected to offer contributions that are truthful, informative, relevant to the goals of the conversation, and appropriately phrased. These expectations about rational conversational conduct constrain the range of interpretations hearers are entitled to entertain in interpreting utterances. Furthermore, these expectations can be violated (or exploited) to create a variety of effects. According to Grice's maxims, in producing (31) and meaning (32), the speaker has used part of the following maxim:

- (35) Quantity Maxim
- i. make your contribution as informative as is required,
 - ii do not make your contribution more informative than is required.
- (Grice 1989: 26)

The speaker has chosen a relatively weak term among a range of words ordered in terms of informational strength: *some* . . . *most* . . . *all*. Assuming that the speaker is trying to be cooperative and will say as much as she truthfully can, the fact that she chose the weaker term *some* gives the listener reason to think that she is not prepared to make the stronger statements in (33) and (34). This leads to the inference that the stronger statement does not hold, that is, to (32). The assumption in (32) is called a conversational implicature, and more specifically, a scalar implicature, since the quantifiers *some* . . . *most* . . . *all* (and the propositions they give rise to) are ordered on a scale (Horn 1972, Gazdar 1979). Implicatures are studied from the perspective of Relevance Theory (Sperber and Wilson 1986) and from a neo-Gricean perspective (Levinson 2000, Chierchia 2004).

Note that conversational implicature is universal, it is purportedly part of human language, and all languages should exhibit a similar process of implied meaning inferencing. Therefore, the issue of transfer from the native language plays out in an interesting way in this area of linguistic pragmatics. The mechanisms of scalar implicature computation, whatever they are, can readily be transferred from the native language of the learner. On the other hand, implicature in certain situations certainly depends on the lexical knowledge of set expressions, or chunks.

A pioneering series of studies on knowledge of conventional implicature was Bouton (1992, 1994). Initially based on a cross-sectional picture, Bouton followed the development of several types of conversational implicature such as relevance and implied criticism. He tested two groups of students after 17 months and after 54 months of residence in the US. The general conclusion from his findings was that the learners were capable of computing implicature after a period of study in the US. The only area of uncertainty and difficulty remained “specific points of American culture and not the type of implicature involved” (Bouton 1994: 163). Bouton’s findings confirmed that implicature is a cognitive process distinct from cultural knowledge and that its acquisition does benefit from instruction and longer exposure to the target language.

There is an extensive literature on scalar implicature computation demonstrating that children can answer experimental questions in a pragmatic way, but not until the ages of 5 to 7 (e.g., Guasti et al. 2005, Papafragou and Musolino 2003). A central question of the child acquisition literature is whether scalar implicature computation development depends on the maturation of some cognitive capacity or on processing abilities. If scalar implicature calculation depends on the maturation of some cognitive capacity in children, we expect adult learners to be much better at it than children learning their mother tongue. Not only are they cognitively mature individuals but their native language is in a position to assist them in inference calculation. If, on the other hand, scalar implicature computation depends on processing capacity because it involves choice of an optimal competitor within a narrow set of options (a sentence with *some* versus a sentence with *not all*, as proposed by Reinhart 2006), we could expect adult L2 learners to have difficulty similar to that of young native speakers. To anticipate the upcoming discussion, since adult L2ers were also revealed to experience difficulty with scalar implicatures, the implication for L1 acquisition is that children's less-than-perfect implicature computation does depend on processing resources.

These two predictions were tested in Slabakova (2010), Snape and Hosoi (to appear), and Lieberman (2009). Slabakova investigated the L2 acquisition of scalar implicatures by Korean-native learners of English. In one experiment the participants had to judge the felicity of underinformative sentences without context and had to say whether they agree with the statement as in (36).

(36) Some elephants have trunks.

A positive answer represents the logical option since *some* and indeed *all* elephants have trunks. However, the sentence is pragmatically infelicitous in that it is not maximally informative; the negative answer is the pragmatic answer. The test sentences were translated in Korean and administered to Korean native speakers, as well as to English natives in English. Slabakova found differences in the Korean speakers' performance in their native and in their second language. They gave around 40% pragmatic answers in their native language (not significantly different from the English native group) and about 60% pragmatic answers in their second language. The results suggest that L2 learners have no problem computing scalar implicatures;

indeed they did so more often than native speakers. In the second experiment with added context, the learners gave pragmatic answers over 90% of the time. Slabakova (2010) argued that the difference between native and second language speakers was due to scalar computations being taxing on the processing resources. Since the logical responses were arguably due to conjuring up alternative contexts in order to agree with the logical use of *some* (e.g., only some elephants have trunks because some others may have been injured or born without trunks¹⁰), speakers had a harder time coming up with these alternative contexts in their second language.

Lieberman (2009) continued the investigation of scalar implicature computation, focusing on the issue of processing resources. He tested the acceptance of computationally demanding implicatures as in (37) and compared them to less demanding sentences as in (38).

(37) Max didn't read all of the books.

(38) Max read some of the books.

A sentence such as (37) involves an indirect implicature because of a scale reversal and is harder to process than the direct implicature in (38), even for native speakers (Gillingham 2007). Lieberman tested Japanese-native learners of English on the scales *<sometimes, always>*, *<partly, completely>*, as well as *every* in the scope of negation. Participants had to evaluate the felicity of sentences in short contexts. When forced to judge the acceptability of single test sentences, both native speakers and learners had difficulty computing the indirect implicatures as compared to the direct ones. The nonnative speakers were even less accurate than the natives, suggesting that in these cases there is indeed a processing problem and the native–nonnative differences are a matter of degree. When the processing load was reduced by presenting the participants with two alternatives to choose from, one felicitous and one infelicitous, the nonnative speakers had no trouble with the task and performed similarly to the native speakers. It is interesting to note that neither in Slabakova (2010) nor in Lieberman (2009) was proficiency a factor in the learners' performance.

A very interesting dimension of child–adult comparisons and processing resources is highlighted by studying bilingual children. There is a well-

¹⁰ As proposed in Guasti et al. (2005).

established effect of bilingualism on executive functioning (involving attention, inhibition, and focusing) in children and adults that we have already discussed on several occasions. Bialystok (2001), Bialystok and Senman (2004), Bialystok and Martin (2004) and others have shown that bilinguals often exhibit significantly superior executive functioning and attentional abilities that are associated with better responses on metacognitive and metalinguistic tasks. Thus, it is possible that the bilingual advantage is a factor in pragmatic development.

This research question was examined by Siegal, Iozzi, and Surian (2009), which compared pragmatic competence in bilingual and monolingual children. Children participating in this study were bilingual in Italian and Slovenian, or monolingual in either language. The researchers tested children aged 3–6 on a conversational violations test to find out whether they would obey Gricean maxims. Results of two experiments in Siegal et al. (2009) show that the bilingual children have a definite advantage over the monolingual ones on four Gricean maxims: Quantity II, Quality, Relation, and Politeness. Bilingual children were more accurate in choosing nonredundant answers, true answers over false ones, answers that were relevant to the questions, and polite answers over rude ones. The only maxim on which all the children performed equally well and hovered at around 60% pragmatic responses was the Maxim of Quantity I. Here is a test item:

(39) Question: “What did you get for your birthday?”

Logical but underinformative answer: “A present.”

Pragmatically appropriate answer: “A bicycle.”

(Siegal et al. 2009: 116)

Results of 60% pragmatic answers for children before the age of 6 are largely in line with other studies in the literature on scalar implicature computation in children. More importantly, however, Siegal et al. (2009) did not establish an advantage for bilingual children comprehending underinformative sentences. Thus it is possible that comprehending underinformative sentences involves different semantic–pragmatic calculations than detecting relevance and rudeness.

In this section, studies were summarized investigating conversational implicature knowledge and scalar implicature knowledge in L2 speakers. Findings suggest that when universal computation mechanisms are at play, learners have no trouble comprehending them; when culturally specific

knowledge or formulaic expressions are involved, learners are less accurate. In addition, the bilingual advantage may only be afforded with respect to the latter but not the former.

11.7 Conclusion

In this chapter, we examined findings from the L2 acquisition of the syntax–discourse and the semantics–pragmatics interfaces. As in other areas of the grammar, there is much that is universal at these interfaces, but there are also distinctions between languages that can be viewed as parametrically opposed. The hypothesis that the interface between syntax and discourse is especially difficult was entertained, as proposed by Sorace’s Interface Hypothesis. However, findings of the studies we reviewed suggested that the syntax–discourse interface is not monolithic in terms of difficulty. In this respect, the syntax–discourse interface resembles other areas of the grammar such as morphosyntax and the lexicon–syntax interface. We saw that even very similar properties across various languages (such as clitic left dislocation in Spanish/Bulgarian and topicalization in English) may create differential problems for learners. Why would this be the case? And does this interface really create difficulties across the board? There are clearly other factors impacting behavior at this interface, such as frequency of the constructions in the input and processing strategies. In addition, even though the semantics–pragmatics interface is largely universal, whether speakers have to expend a lot of mental energy or not for calculating a meaning at this interface is another important factor.

11.8 Exercises

Exercise 11.1. This is a question on a methodological issue. In our discussion of second language acquisition, we can in principle compare the accurate comprehension or production of native and nonnative, or even near-native speakers of some language *L*. Imagine that we have created an experiment on some construction *X*, and we have tested native and nonnative speakers. We have included acceptable and unacceptable sentences in our test. Figures 11.3 and 11.4 are hypothetical, but such results are all too often attested in real experiments.

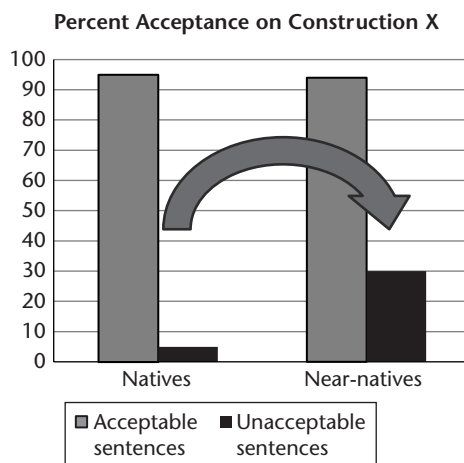


Figure 11.3 Hypothetical research result situation. The arrow indicates the group means that are not informative to compare: whether or not native speakers and learners reject an unacceptable construction with equal accuracy.

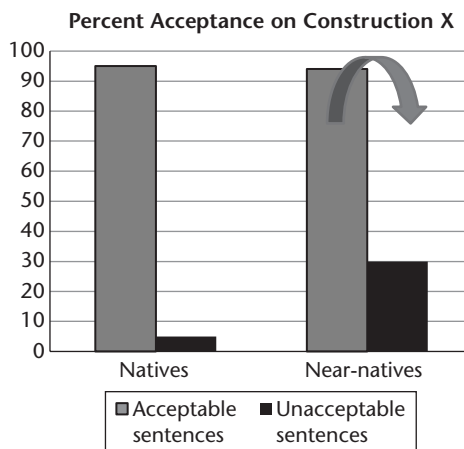


Figure 11.4 Hypothetical research result situation. The arrow indicates the group means that are informative to compare: whether or not the learners distinguish reliably between unacceptable and acceptable test items within the same construction.

What should we compare in such a situation, when the acceptance rates are very similar but the rejection rates differ? Should we compare the acceptance rates of the two constructions across speaker groups, natives and near-natives? Or should we be asking ourselves if the near-natives distinguish reliably between these two types of sentences? What is more important: that

learners are indistinguishable from native speakers, or that learners acquire the contrasts in the target grammar? This choice refers to the Comparative Fallacy, proposed by Bley-Vroman (1983) and discussed in Chapter 10. Find the publication and discuss its main point. Then answer the questions I asked with respect to the two graphs above.

Exercise 11.2. Create experimental items. Here are two examples of sentence-story combinations from Slabakova and García Mayo (2015). The idea is to check whether learners of English know that focus fronting and topicalization involve moving an object to the beginning of the clause without a resumptive pronoun in the object position. In order to check on that knowledge, the experimental design offers a general context and an immediate context. Sentences are recorded to ensure the correct intonation. Participants are asked to evaluate the acceptability of the last sentence in the light of the story and the immediate context.

Task: Create three experimental items of each sort. Make sure to provide two dialogs for each, one with a resumptive pronoun and one without. The order of presentation of the acceptable and unacceptable sentence under each context has to be randomized.

(i) Fronted Focus

Context: Mark works in a bakery in San Francisco. Their bagels are so famous that they often run out of them by the middle of the morning. This has happened again today.

Dialog 1

Customer: Can I get something for breakfast? Maybe a bagel?

Mark: No, sorry. We're out of bagels. A bran muffin I can give you.

Dialog 2

Customer: Can I get something for breakfast? Maybe a bagel?

Mark: No, sorry. We're out of bagels. A bran muffin I can give it to you.

(ii) Topicalization

Context: John and his sister Sophie are in a Japanese restaurant. John has never eaten in this restaurant before, so he is not sure about what to order. He decides to ask Sophie.

Dialog 1

John: Have you tried the fish here?

Sophie: Last week I had the sole. It was delicious. The salmon I haven't tried it yet.

Dialog 2

John: Have you tried the fish here?

Sophie: Last week I had the sole. It was delicious. The salmon I haven't tried yet.

Question: It is also possible to present each general context followed by one dialog, and then the same context followed by the other dialog, but in another place in the test.¹¹ Discuss the advantages and disadvantages of the two presentation modes. (Hint: One is better but the test takes more time.)

Exercise 11.3. Likert scales. The experiment mentioned in the previous exercise uses a Likert scale from 1 to 7 as shown below, where 1 stands for “the test sentence does not sound natural in this context,” while 7 stands for “the test sentence sounds perfectly natural in this context.” There is a separate point for IDK (“I don’t know”).

1-----	2-----	3-----	4-----	5-----	6-----	7	I don't know
Not						Perfectly	
natural						natural	

What is a Likert scale? And how do we pronounce the name Likert? Is the first syllable “like” or “lick”? While both pronunciations are attested, Americans seem to prefer Like-ert, while Europeans tend towards Lick-ert. The correct pronunciation is with a short [ɪ] in the first syllable. Rensis Likert (1903–81) was an American administrator and organizational psychologist. His most famous creation is the eponymously named scale, a psychometric scale commonly involved in research using questionnaires. He proposed it in his dissertation and published the paper in 1932 (Likert 1932). It is used for measuring attitudes and is argued to capture more information than competing methods. The traditional Likert scale asks people the extent to which they agree or disagree with a statement on a 5-point scale. The scale ranges from 1=strongly agree, 2=agree, 3=undecided/neutral,

¹¹ Randomization is done, for example, by generating random numbers and entering each one of them in front of every story, then sorting. It is also a good idea to do pseudo-randomization, mixing the sentences up by blocks, so that similar conditions do not end up close together. Another option is to use survey services such as SurveyGizmo, Qualtrics, or Survey Monkey, which randomize the test item presentation for each participant.

4=disagree, 5=strongly disagree. Using a 5-point scale allowed Likert to rank people's attitudes with fewer questions and greater exactness. Notice that there is no true zero in this scale.

In applied linguistics, we use the Likert scale when we expect participants' judgments to be subtle and have nuances. Otherwise, we use a binary acceptability choice: Yes or No. Why do you think Slabakova and Garcia Mayo used a scale with 7 points? What would they have lost if the choices were simply Yes or No? Find the article and read the methodological discussion of the scale.

Why is it important that there is a separate answer point for IDK? Research the classical Likert scale and compare it with the usage in applied linguistics.

Read this publication by JD Brown, a well-known specialist on research methods and statistics in applied linguistics:

HTML: http://jalt.org/test/bro_34.htm

PDF: <http://jalt.org/test/PDF/Brown34.pdf>

Exercise 11.4. Roberts, Gullberg, and Indefrey (2008).

While this exercise is based on a published study, do not look at the original study before going through all the questions and attempting answers. After you are done with this exercise, allow yourself the pleasure to look at the study and compare your interpretations. First, let's look at the linguistic facts.

- (1) a. English:
Peter and Hans are in the office. While Peter is working, he is eating a sandwich.
- b. Dutch: Unstressed pronoun
Peter en Hans zitten in het kantoor. Terwijl Peter₁ aan het werk is, eet hij₁ een boterham.
- c. Turkish: Zero pronoun
Peter ve Hans₂ ofiste oturuyorlar. Peter₁ çalışırken, Ø_{1/*2} sandeвиç yiyor.
- d. Turkish: Overt subject pronoun
Peter ve Hans₂ ofiste oturuyorlar. Peter₁ çalışırken, o_{*1/2} sandeвиç yiyor.

Question 1: Describe the linguistic facts as you see them in the English, Dutch, and Turkish examples in (1). First of all, who does *he* refer to in the English sentence? Is Peter the only possible antecedent, or the preferred antecedent? Note that the sentences involve the same constituents, although Turkish and Dutch are SOV languages. If a pronoun and a name share the same subscript (e.g., 1) they refer to the same individual, for example *Peter* and *hij* in (1b). $\emptyset$ stands for the null pronoun while *o* is the Turkish equivalent of *he*. If a pronoun has two indexes 1/2, then it is ambiguous because it can refer to two possible antecedents. Which two languages of these three work similarly? Is there a reason for that?

Question 2: Roberts et al. (2008) tested three groups of participants: native speakers of Dutch and German and Turkish learners of Dutch at advanced proficiency. Can you explain the choice of these languages?

All participants undertook three experimental tasks, in which exactly the same experimental items appeared, to allow for a better comparison of the results across tasks. The session began with an eye-tracking experiment that was designed in such a way that the texts were read purely for meaning. Then an offline acceptability judgment task was administered, during which participants read each text and were asked to rate each one on a scale from 1 (least acceptable) to 6 (most acceptable). Finally, the participants completed a comprehension questionnaire. This was chosen as the last task because it specifically probed the participants' preferred referent for the subject pronoun, whereas the other two tasks did not.

Here are some sample test items, examine them all carefully. The first sentence introduces the context with an antecedent. It happens to be an antecedent external to the second, and all-important, test sentence. That sentence also introduces an antecedent, which we will call the local antecedent because it is in the same sentence as the pronoun in bold. Finally, the third sentence rounds off the context.

- (2) a. De werknemers zitten in het kantoor. Terwijl Peter aan het werk is, eet **hij** een boterham. Het is een rustige dag.
'The workers are in the office. While Peter is working, he is eating a sandwich. It is a quiet day.'
- b. De werknemers zitten in het kantoor. Terwijl Peter aan het werk is, eten **zij** een boterham. Het is een rustige dag.
'The workers are in the office. While Peter is working, they are eating a sandwich. It is a quiet day.'

- c. Peter en Hans zitten in het kantoor. Terwijl Peter aan het werk is, eet **hij** een boterham. Het is een rustige dag.
 ‘Peter and Hans are in the office. While Peter is working, he is eating a sandwich. It is a quiet day.’

Question 3: Identify the intended antecedent of the Dutch pronouns in bold, dependent on the context. Can you predict which of these sentences would be easiest to process, and why? Consider both the ambiguity of the pronoun and Topic shift.

We will summarize below some of the findings of the three experimental tasks. Let us call (2a) the *local* antecedent condition, (2b) the *external* antecedent condition, and (2c) the *ambiguous* condition because the two antecedents can be chosen optionally. Remember that all conditions were present in all three tasks.

In the **acceptability judgment** task, all groups performed similarly. Both natives and learners rated the external antecedent condition slightly lower than the other two, but found them largely acceptable.

The **comprehension** task actually asked for the speakers’ preferred antecedent. The results are given in Table 11.1.

Question 4: Discuss these results. First of all, do you see any optionality in the behavior of the Dutch natives, in the “optional antecedent” condition? Which group seems to be choosing antecedents differently from the rest? Why do you think that is the case? Can we attribute it to L1 influence?

The results of the eye-tracking during reading task are quite complicated. The assumption underlying this task was that the time spent reading the region in which the pronoun antecedent is decided (also known as pronoun resolution) is a measure of how accessible the referent is and, thus, of the comparative difficulty of the pronoun resolution process. Simplistically put, the longer a reader gazes at a pronoun, the harder she is thinking about which is its possible antecedent. Both the Turkish and the German L2

Table 11.1 Mean (SDs) percentage of sentence internal referent (or local antecedent) chosen for the pronoun in the embedded clause of the second sentence

Group	Local antecedent (2a)	External antecedent (2b)	Optional antecedent (2c)
Dutch	100 (0)	7 (28)	100 (0)
German	98 (6)	8 (15)	91 (18)
Turkish	95 (14)	0 (0)	55 (33)

learners spent **more time** reading the critical region in the optional condition than in the local and external antecedent conditions. They also returned to gaze at the pronoun more times in that condition (compared to the other two).

Question 5: What can we make of these disparate results? How can we interpret the behavior of the bilinguals? You may want to assume, as the authors of the study do, that the acceptability judgment task is the most metalinguistic of the tasks, in the sense that the learners can think of rules they have been taught. The comprehension task is most reflective of underlying competence with respect to pronoun antecedent choice in discourse, while the eye-tracking task reflects processing as well as competence. Keep in mind that German and Dutch work similarly while Turkish (being a null-subject language) is like the Spanish and Italian examples we saw in Section 11.3.